

1. Context of Testing

1.1 Project/Test Levels/Test Types

- **Project:** Functional and non-functional testing of the Trendyol web and mobile e-commerce platform.
- **Test Levels:** System testing and acceptance testing.
- **Test Types:** Functional testing, boundary value testing, equivalence class testing, decision table testing.

1.2 Test Items

- **Items:** Login/sign up email field, product search bar, shipping fee logic, placing an order process.
- **Version:** As accessed in May–June 2025 from <https://www.trendyol.com>.

1.3 Test Scope

- **Features to be Tested:**
 - Email input validation in login/sign up (Equivalence Class Partitioning)
 - Product search bar edge input handling (Boundary Value Analysis)
 - Shipping fee rules and conditions (Decision Table Testing)
 - Placing an order from product page to confirmation (Use Case Testing)
- **Features Excluded:**
 - Internal admin panel
 - Third-party integrations
 - In-app messaging system

1.4 Test Basis

- Application user behavior
- Functional specifications inferred from the live application
- Common e-commerce use cases such as sign-up, search, checkout, and payment

2. Assumptions and Constraints

- No access to source code (black-box testing only)
- Testing is performed on publicly available production version
- Limited test data due to lack of access to backend
- All group members are students with limited automation experience

3. Test Strategy

3.1 General

- Manual testing for exploratory and initial verification

- Automated testing for regression and critical functionality

3.2 Test Levels

- System-level testing for end-to-end user flows
- Acceptance-level testing to simulate user scenarios

3.3 Test Types

- Functional
- Boundary value (Search Bar)
- Equivalence partitioning (Email field)
- Decision table testing (Shipping Fee)
- Use Case testing (Place an Order)
- Automation (Selenium IDE + JUnit)

3.4 Test Deliverables

- Test Plan (this document)
- Test Case Specification
- Selenium IDE Scripts
- JUnit-based Automation Scripts
- Test Completion Report
- Presentation Slides

3.5 Test Design Techniques

- Equivalence Class Partitioning (Sign up/Login email input)
- Boundary Value Analysis (Search bar input length)
- Decision Tables (Shipping fee based on cart total and promotions)
- Use Case Testing (Complete order placement scenario)

3.6 Entry and Exit Criteria

- **Entry:**
 - Functional test cases defined and reviewed
 - Selenium scripts partially working
 - Test data prepared
- **Exit:**
 - All major functionality covered with at least 1 test case
 - All critical bugs logged and resolved or accepted

3.7 Test Completion Criteria

- All high-priority test cases executed
- All defined use cases tested
- Test summary reviewed and approved by team

3.8 Metrics to be Collected

- Total test cases
- Passed/failed test cases
- Time taken per test case
- Coverage (manual vs automated)

3.9 Test Data Requirements

- Valid/invalid user accounts for login/signup
- Valid/invalid email formats
- Search keywords (valid terms, empty, max-length)
- Cart totals triggering free shipping or not
- Products with varying stock availability

3.10 Test Environment Requirements

- Browsers: Chrome, Firefox (latest versions)
- Devices: Android phone, iPhone
- Tools: Selenium IDE, IntelliJ IDEA, JUnit, Excel

3.11 Testing Activities and Estimates

- Manual testing: 20 hours
- Selenium IDE setup and automation: 10 hours
- JUnit code automation: 10 hours
- Documentation and reporting: 8 hours

4. Staffing

4.1 General

- 4 students will equally participate in test planning, execution, automation, and reporting.

4.2 Roles and Responsibilities

Name	Primary Role	Secondary Role
Arda Gürkan	Unit Test Developer	Manual Test Executor
Can Gere	JUnit Test Developer	JUnit Test Developer
Emir Ersan	Selenium IDE Automation	Report Writer
Tibet Derebağ	Selenium IDE Automation	Test Execution Tracker

4.3 Schedule

Activity	Dates
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Test Planning	May 25
Manual Testing	May 28–June 5
Automation (IDE & Code)	June 6–June 13
Reporting and Finalizing	June 14–June 18
Presentation Preparation	June 17–June 19
Presentation	June 20
